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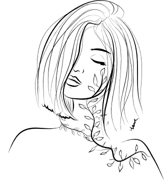


Q1: HOW ARE OSI AND ISO RELATED TO EACH OTHER?

The OSI (Open Systems Interconnection) model is a conceptual framework for understanding . network communication, developed by ISO (International Organization for Standardization) in 1974

- ISO is the organization that developed and standardized the OSI model

Q2: MATCH THE FOLLOWING TO ONE OR MORE LAYERS OF THE OSI MODEL:

a. Route determination → Network Layer

b. Flow control → Transport Layer and Data Link Layer

c. Interface to transmission media → Physical Layer

d. Provides access for the end user → Application Layer

Q3: MATCH THE FOLLOWING TO ONE OR MORE LAYERS OF THE OSI MODEL:

a. Reliable process-to-process message delivery → Transport Layer

b. Route selection → Network Layer

c. Defines frames → Data Link Layer

d. Provides user services such as e-mail and file transfer → Application Layer

e. Transmission of bit stream across physical medium → Physical Layer

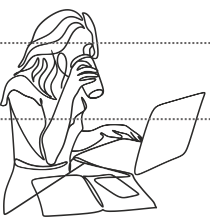
Q4: MATCH THE FOLLOWING TO ONE OR MORE LAYERS OF THE OSI MODEL:

a. Communicates directly with user's application program → Application Layer

b. Error correction and retransmission → Data Link Layer

c. Mechanical, electrical, and functional interface → Physical Layer

d. Responsibility for carrying frames between adjacent nodes → Data Link Layer



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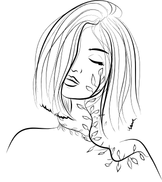


Q5: MATCH THE FOLLOWING TO ONE OR MORE LAYERS OF THE OSI MODEL:

a. Format and code conversion services → Presentation Layer

b. Establishes, manages, and terminates sessions → Session Layer

c. Ensures reliable transmission of data → Transport Layer

d. Log-in and log-out procedures → Session Layer

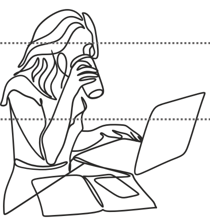
e. Provides independence from differences in data representation → Presentation Layer

Q6: SUPPOSE A COMPUTER SENDS A PACKET AT THE NETWORK LAYER TO ANOTHER COMPUTER SOMEWHERE IN THE INTERNET. THE LOGICAL DESTINATION ADDRESS OF THE PACKET IS CORRUPTED. WHAT HAPPENS TO THE PACKET? HOW CAN THE SOURCE COMPUTER BE INFORMED OF THE SITUATION?

- Since the packet has a corrupted logical (IP) address, it will likely be dropped by the router or the destination network because it cannot be correctly routed.
- The source computer can be informed of the situation by the Internet Control Message Protocol (ICMP) which sends error messages back to the sender, such as a "Destination Unreachable" message.

Q7: SUPPOSE A COMPUTER SENDS A PACKET AT THE TRANSPORT LAYER TO ANOTHER COMPUTER SOMEWHERE IN THE INTERNET. THERE IS NO PROCESS WITH THE DESTINATION PORT ADDRESS RUNNING AT THE DESTINATION COMPUTER. WHAT WILL HAPPEN?

- If no process is listening on the destination port, the destination computer will send a message back to the sender indicating that the port is unreachable.
- This is typically done using the ICMP "Port Unreachable" message or a TCP reset (RST) packet if TCP is used.



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Q8: IN FIGURE 1, COMPUTER A SENDS A MESSAGE TO COMPUTER D VIA LAN 1, ROUTER R1, AND LAN 2. SHOW THE CONTENTS OF THE PACKETS AND FRAMES AT THE NETWORK AND DATA LINK LAYER FOR EACH HOP INTERFACE.

At LAN 1 (Computer A to Router R1):

- Network Layer: Packet with source IP = A's IP, destination IP = D's IP.
- Data Link Layer: Frame with source MAC = A's MAC, destination MAC = R1's MAC on LAN 1.

At Router R1 (between LAN 1 and LAN 2):

- Router strips the frame from LAN 1 and creates a new frame for LAN 2.
- Network Layer: Packet remains the same (source IP = A, destination IP = D).
- Data Link Layer: Frame with source MAC = R1's MAC on LAN 2, destination MAC = D's MAC.

At LAN 2 (Router R1 to Computer D):

- Frame is delivered to D with correct MAC addresses.
- Packet is forwarded unchanged at the network layer.